Shivam Kamboj — Publication List

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Peer-reviewed publication

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, "Oscillatory edge modes in two-dimensional spin-torque oscillator arrays," *Physical Review B* **109**, 094436 (2024). doi:10.1103/PhysRevB.109.094436

Manuscripts in preparation

2. S. Kamboj and L. Tian, "Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer."

3. S. Kamboj, J. Beurkens, J. Sharping, and L. Tian, "Superconducting realization of the Quantum Rabi Dimer."

Open-source research software

4. S. Kamboj, `nonmarkov_qec`: surface-code benchmarking under temporally correlated dephasing (Stim, PyMatching, Monte Carlo sampling with bootstrap uncertainties). Released on PyPI, 2026. github.com/shivam1kamboj/nonmarkov_qec

Presentations

- Quantum Industry Showcase, UC Santa Barbara, 2026
  Author: Shivam Kamboj, Lin Tian
  Title: The Instability Window: Macroscopic Entanglement and Delayed Symmetry Breaking in the Quantum Rabi Dimer

- APS Global Physics Summit, 2026.
  Author: Shivam Kamboj, Lin Tian
  Title: Adiabatic Evolution and Fock-state Mapping Across the Phase Boundary in a Quantum Rabi Dimer

- Poster, PLS Division Student Poster Symposium, Lawrence Livermore National Laboratory, 2024.
  Author: Shivam Kamboj, Yujin Cho, Yaniv J. Rosen
  Title: Improving Fidelity of Quantum Gates Using Optimal Control

- Invited talk, Friday Seminar Series, CSU Stanislaus, 2024.
  Author: Shivam Kamboj
  Title: Simulating Quantum Systems, and Quantum Key distribution

- APS March Meeting, Chicago, 2022. Abstract T00.249.
  Author: S. Kamboj and H. M. Hurst
  Title: Topology of a Driven 2D Spin Torque Oscillator Array